

BTS (C) 048 (E)

B.Tech. Degree IV Semester Examination
June 2002

ME 403 THERMAL ENGINEERING - II
(2000 Admissions)

Time: 3 Hours

Max.Marks:100

- I a) A multistage steam turbine has two stages of feed heating; the temperature rise of feed water is to be equally divided between two heaters. Initial steam pressure is 30 bar with temperature 320°C . The exhaust pressure is 7 kN/m^2 and the temperature of condensate is 37°C . The first tapping is made at 2 bar. Calculate the pressure for the second tapping point and the qualities to be tapped at each point. Also calculate the gain in efficiency due to feed heating and increase in consumption rate. Take stage efficiency as 0.7 and consider feed heaters as closed one. }
b) What are the advantages of regenerative heating in Rankine cycle? }20

OR

II a) Derive an expression for critical pressure ratio in steam nozzles. (10)
b) Superheated steam enters a nozzle at a pressure of 25 bar and temperature 300°C and expands adiabatically to an exit pressure of 2 bar. Calculate
(i) exit speed of steam neglecting initial speed of steam at inlet of nozzle.
(ii) exit area of nozzle for a mass flow rate of steam 1.25 kg/s . (10)

III a) Derive an expression for speed ratio corresponding to maximum blade efficiency in steam turbine. (10)

(P.T.O.)

- b) An impulse steam turbine has exit velocity of steam from a converging nozzle equal to 900 m/s. The nozzle makes an angle of 20° to the tangent of rotating wheel. The mean blade speed is 300 m/s. the blades are equiangular. the mass of steam flowing through the turbine is 1200 kg/hr. Calculate
 (i) the blade angles
 (ii) axial thrust on blades
 (iii) power developed
 Take blade velocity coefficient $k = 0.85$. (10)
OR
- IV a) Explain a gas turbine cycle with inter cooling, reheating and regeneration. (8)
- b) A gas turbine operates between pressure limits of 100 kN/m² and 500 kN/m². The inlet air temperature to the compressor is 15°C and the gas entering the turbine is at a temperature of 537°C . If the volume of air entering the compressor is 1400 m³/mt., calculate the net available power output for the cycle and the efficiency. $C_{p,\text{air}} = 1.005 \text{ KJ/KgK}$, $C_{p,\text{gas}} = 1.15 \text{ KJ/KgK}$ and for both air and gas $\gamma = 1.4$. (12)
- V a) An ideal vapour refrigeration cycle uses Freon-12 as the working fluid. The condensing temperature is 38°C and the evaporation temperature is -7°C . Calculate
 (i) COP of cycle
 (ii) power required to produce 1 ton of refrigeration
 (iii) Circulation rate of fluid for each ton of refrigeration. (10)
- b) Explain cooling systems for air craft cooling. (10)
OR
- VI a) Compare properties of different refrigerants. (7)
- b) An air refrigeration system operating on Bell-Coleman cycle takes in air from cold room at -5°C and compresses it from cold room at -5°C and compress it from 1.0 bar to 5.5 bar. Index of compression being 1.25. The compressed air is cooled to 27°C . Ambient temperature is 20°C . Air expands in an expander where index of expansion is 1.35. (13)

- VII a) Explain about different psychrometric processes. (7)
- b) An office space is to be air conditioned for 60 people. The outdoor conditions are DBT = 30°C , RH = 60% Quantity of air supplied is 0.4 m³/mt/person. Determine: (i) Capacity of cooling coil in kw.
 (ii) Capacity of heating coil in kw
 (iii) Amount of water vapour removed per hour
 Assume that
 (1) Inlet air has 60°C DBT and 60% RH
 (2) Air is conditioned first by cooling and dehumidifying and then by heating. (13)
OR
- VIII. a) Explain about sensible heat factor. (7)
- b) 1000 cubic meters of air per minute at 35°C DBT and 60%RH is cooled to 20°C DBT by passing through the cooling coil. Find the following:
 (i) Capacity of cooling coil in KJ/hr.
 (ii) Amount of water vapour removed/hr.
 (iii) RH of air coming out and its WBT. (13)
- IX a) Draw a layout of thermal power plant. Explain function of different components. (13)
- b) Explain about different dust handling and collection systems. (7)
OR
- X a) With a neat sketch explain functions of various components of nuclear reactor. (10)
- b) What is meant by a fast breeder reactor? Explain differences between FBR and BWR. (10)

